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Can donors prevent aid misallocations? Evidence from Chinese and World Bank aid☆

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ABSTRACT

We estimate the extent to which international aid projects are subject to favoritism using differences in differences and RDD estimates based on presidential transition dates. We compare two different sources of aid, China and the World Bank, and two different types of favoritism based on the president's region of birth or ethnic group. Consistent with the literature, we find that Chinese aid is massively targeted to the region of birth of new presidents. However, we also find evidence of a less visible and less intense form of favoritism in World Bank aid: it targets areas that are co-ethnic with a new president, without following major regional administrative boundaries. This is consistent with a better control of the World Bank aid. Moreover, the political economy of World Bank aid favoritism appears to be more complex: World Bank aid favoritism is more intense in dictatorships, when Chinese aid is more abundant, and in countries that have historically received more World Bank aid.

1. Introduction

The effectiveness of aid is one of the great controversies among development economists. Aid can sometimes increase the well-being of local populations tremendously (see Burnside & Dollar, 2000; Cruzatti, Dreher, & Matzat, 2023; Dreher, Fuchs, Hodler, &, Tierney, 2021; Dreher, Fuchs, Parks, Strange, & Tierney, 2021; Dreher, Lang, & Reinsberg, 2024; Svensson, 1999). However, the political economy of aid explains some of the skepticism: aid can be politically motivated by donors (Alesina & Dollar, 2000), can alter the political equilibrium of recipient countries (Djankov, Montalvo, & Reynal-Querol, 2008), and can be captured by recipient country elites (Jablonski, 2014).

This paper questions the ability of different types of donors to prevent the capture of aid for political purposes. We first compare multilateral aid from the World Bank with bilateral aid from China, which has been very active in Africa in recent years. Multilateral aid often has more explicit development objectives than bilateral aid, which is likely to be more responsive to political and commercial objectives and thus less subject to scrutiny.[1] We also examine whether the context in which aid is provided affects the possibility of capture. We first examine the institutional context, where better institutions are likely to allow for better control of aid. We also examine the degree of competition among donors, with greater competition potentially leading to less donor control and the role of pre-existing relationships in aid between the donor and the recipient. To this end, we simultaneously study presidential birthplace favoritism and co-ethnic regions favoritism for both Chinese and World Bank aid.[2] We argue that ethnic favoritism targets a very different set of zones than the president's birthplace and is less visible. It does not follow the official internal borders of the country, and ethnic regions are less compact: ethnic favoritism simultaneously targets many

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[1] Multilateral donors include the World Bank, the European Union, and the United Nations, and account for 25% of development aid. Together with the EU, the World Bank is the largest multilateral donor. Multilateral aid likely to be less captured by donors' political considerations, as multilateral agencies represent several countries. The World Bank has a long history of monitoring and evaluating its activities and promoting transparency. We explain how the World Bank's control policies are more developed than China's in Appendix B.

[2] The literature generally identifies two forms of favoritism in African politics. Presidents can distort public policy in favor of their birthplace (see Bandyopadhyay and Green 2019, Dreher et al. 2019, Hodler and Raschky 2014) or in favor of coethnic regions (see Burgess, Jedwab, Miguel, Morjaria, and Padró i Miquel 2015, Franck and Rainer 2012, Kramon and Posner 2016 or Dickens 2018).

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places with less clear borders. Birthplace favoritism may therefore be more visible and consequently more difficult to implement if donors want to limit favoritism and exert more control over aid. Overall, it is useful to study these two dimensions separately, as they overlap only slightly.[3]

Using data on Chinese and World Bank aid to Africa between 1995 and 2014, we estimate the effect of geographic favoritism on aid allocation with difference-in-differences specifications and regression discontinuity designs.[4] We find strong evidence of differential targeting for Chinese and World Bank aid: Chinese aid is subject to birth region targeting (as previously shown by Dreher et al., 2019), and World Bank aid appears to be mostly subject to ethnic favoritism. When a president changes, Chinese aid to her birth region increases sharply (taking other regions as a reference). The magnitude of the effect is impressive and much larger than previously measured in the literature: Chinese aid to a new president's region of birth is multiplied by about 7 in the decade following her appointment, relative to other regions. We argue that methodological differences may render previous estimates somewhat conservative.[5]

World Bank aid follows a very different pattern. When a new president takes office, World Bank aid increases mainly in the ADM2 districts of her ethnic group. The magnitude is much smaller, but still substantial: the amounts roughly double. This finding is relatively new in the literature. Dreher et al. (2019) find no geographic favoritism based on place of birth or ethnicity associated with World Bank aid. A notable exception is Perrotta Berlin, Desai, and ard (2023) who also study geographic favoritism based on ethnicity for World Bank aid but find that co-ethnic regions receive less aid from the World Bank. However, they have a different focus, studying the effect of having a permanent seat in the UN on aid allocation within the country. Moreover, their sample covers the whole world, while we focus on Africa, where ethnicity is most salient. We also highlight the role of institutions, donor competition and pre-existing relationships in explaining favoritism with respect to these two papers.[6]

The birth region of the newly appointed president accounts for 20% of Chinese aid on average. We estimate that 18 percentage points (90%) can be explained by favoritism. The coethnic districts of a newly appointed president account for 4% of World Bank aid, and 1 to 3 percentage points can be explained by favoritism.

This is consistent with the view that favoritism in World Bank aid is probably less visible because the amounts are smaller, but also because ethnic regions are less geographically clustered and do not always follow major administrative boundaries, suggesting better control. In addition, donors have better knowledge of administrative boundaries than of the details of ethnic regions. Also consistent with this view, we find no evidence that World Bank aid follows the ethnicity of administrative regions (ADM1): it follows smaller and less visible districts (ADM2), suggesting fine redistribution within regions.[7]

We argue that this particular pattern of favoritism in aid allocation is likely due to a lower level of control over Chinese aid compared to World Bank aid. This may be because multilateral aid has a less political agenda and is therefore more focused on development objectives, and its use is therefore more controlled than bilateral aid. In practice, the World Bank has detailed anti-corruption guidelines and controls, and an internal team assesses the relevance of projects to development objectives and applies transparency to its activities, making it more difficult to implement projects that are inefficient in terms of development objectives, for example because of favoritism. Such controls are much less developed for Chinese aid (see details in Appendix B). Note, however, that Chinese aid does not necessarily have more political or commercial objectives than bilateral aid from other donors (see Dreher and Fuchs (2015) or Dreher, Fuchs, Parks, Strange, and Tierney (2018)).

Our second contribution concerns the heterogeneity of World Bank and Chinese aid favoritism. First, we find that the degree of favoritism depends on the quality of institutions. Co-ethnic regions receive more aid from the World Bank only during autocratic periods. More specifically, two dimensions of democracy prevent ethnic favoritism in the allocation of World Bank aid: constraints on the executive and the presence of an organized selection process of the executive. This seems consistent: when accountability increases and legal selection processes are respected, politicians have incentives to be less clientelistic and to target a broader spectrum of the population. Surprisingly, favoritism in Chinese aid is observed in both autocracies and democracies. This suggests that control over Chinese aid is likely to be weaker, and that control over aid and institutions may complement each other. To our knowledge, these findings on the role of institutions in aid capture are new to the literature with the notable exception of Bommer, Dreher, and Perez-Alvarez (2022) who show that the quality of institutions matters in explaining the leader's birth region bias in U.S. aid allocation after a disaster.

We also find that competition between donors is likely to be important in explaining the degree of aid favoritism. We find that World Bank aid favoritism disappears when the World Bank did not give to the country before the transition, and is smaller when Chinese aid did not exist. In contrast, Chinese aid favoritism is unaffected by the intensity of World Bank aid. These results suggest a form of donor competition, in that abundant Chinese aid appears to relax control over World Bank aid. There is a recent literature showing significant competition between donors (Asmus, Eichenauer, Fuchs, & Parks, 2023; Watkins, 2022; Zeitz, 2021) but the fact that donor competition affects favoritism, likely through a relaxation of controls is a new finding in this literature. Finally, we find that favoritism in World Bank aid allocation is stronger when there is a pre-existing relationship with the recipient country, which we measure by the amount of aid prior to the change of president. We interpret this finding by suggesting that the World Bank loses some of its bargaining power with national governments when pre-existing relations are more intense, possibly because limiting aid in countries with good relations is costly, for example to some actors within the World Bank.

Our results are robust to a variety of specifications and estimators. Event study analyses confirm the absence of pre-trends and underscore the robustness of our differences- in-differences. RDD explicitly controls for a specific trend before and after transitions, and in treated and untreated regions, and leads to similar (if not quantitatively larger) results.

[3] We find that only 40% of the districts (ADM2) of birth are co-ethnic with the president and that the district of birth represents only and 1% of the co-ethnic districts, see Section 3 for details.

[4] Studying Africa has advantages. First, ethnicity is salient, and most districts (ADM 2 administrative regions) are ethnically homogeneous: the local majority ethnic group represents 62% of the district. This makes ethnic targeting of aid feasible (Ejdemyr, Kramon, & Robinson, 2018) and visible: geographic favoritism is sufficient to achieve ethnic favoritism. Finally, Africa has experienced a relatively recent period of democratization. This provides valuable information on the dependence of favoritism on the institutional environment. On the other hand, this may come at a cost in terms of loss of generality.

[5] We show in the paper that these notable differences in the magnitude of the results are due to the lack of accounting for pre-trends in the previous literature.

[6] Another difference is that they study ADM1 regions, we only observe ethnic favoritism at the finer ADM2 level.

[7] Chinese aid favoritism is visible at the ADM1 (regional) level, but not at the ADM2 (district) level. This is probably due to the different types of projects implemented by China and the World Bank. Chinese aid often corresponds to large infrastructure projects, while the World Bank also finances smaller projects in health or education (targeting smaller zones such as ADM2 districts).

The remainder of the paper is organized as follows. Section 2 reviews the literature on aid and favoritism and details our contributions. Section 3 presents the data and the construction of the main variables. Section 4 presents the empirical strategy and Section 5 the results. Section 6 concludes the paper.

2. Literature review

This paper contributes to the literature on favoritism and pork-barrel politics, an important topic in the field of African politics (André, Maarek, & Tapo, 2018; Burgess et al., 2015, 2015; De Luca, Hodler, Raschky, & Valsecchi, 2017; Dreher et al., 2019; Franck & Rainer, 2012; Hodler & Raschky, 2014; Mueller & Tapsoba, 2016). This literature often studies ethnic favoritism and birthplace favoritism separately, with notable exceptions (Bommer et al., 2022; Dreher et al., 2019; Perrotta Berlin et al., 2023). An important question remains: when and why does the leader favor her region of birth and/or her ethnic group? These two forms of elite capture seem to coexist in the literature and may have very different consequences and political returns. Birth region favoritism targets a more compact region. Ethnic favoritism is more geographically dispersed and targets political supporters, most of whom are treated impersonally: they have no personal ties to the president. Measuring favoritism using aid data has several advantages: Aid data includes the date and location of many projects and is comparable across a wide range of countries.[8]

We also contribute to the literature on the political economy of aid (Alesina & Dollar, 2000; Bräutigam & Knack, 2004; Dreher, Lamla, Lein, & Somogyi, 2009; Rajan & Subramanian, 2007; Reinikka & Svensson, 2004; Svensson, 2000). Aid can be captured for reasons related to donors' objectives and controls,[9] and for reasons related to the political economy of recipients.[10] Donors have very different objectives and different levels of internal controls. Internal controls are arguably better developed at the World Bank, and World Bank aid has explicit development objectives.[11]

Since the turn of the century, new donors such as China have emerged. China was the second largest donor on the continent between 2000 and 2010 (Hernandez, 2017). China has a policy of non-interference in the internal affairs of recipients (Jiabao, 2004), which contrasts with the World Bank's conditionality approach.[12] Chinese aid has been accused of undermining aid conditionality and the fight against corruption (Naim, 2007). The World Bank appears to apply less conditionality in countries receiving Chinese aid (Hernandez, 2017; Watkins, 2022). It should be noted, however, that there is no clear evidence that Chinese aid is more closely aligned with geopolitical or economic interests than other bilateral donors (Dreher & Fuchs, 2015; Dreher, Nunnenkamp, & Thiele, 2011), except for flows not classified as overseas development assistance (Dreher et al., 2018). We find different forms and magnitudes of aid captured from two very different donors. This suggests that donors can at least partially control aid capture without eliminating it entirely.

A recent contribution to this literature suggests some degree of competition between donors. Asmus et al. (2023) show that Indian aid is more abundant in districts where Chinese aid was observed in the previous year. Zeitz (2021) shows that the sectoral composition of World Bank aid has changed with China's rise in international aid, shifting towards sectors in which China is active, such as infrastructure. Watkins (2022) shows that the presence of Chinese aid reduces compliance with World Bank aid conditionality by providing an outside option. Our argument is similar to that of Watkins (2022): we show that competition between donors can potentially increase aid favoritism, presumably by relaxing controls.

This paper is also closely related to the intersection of favoritism and aid allocation. Using fixed effects and OLS specifications, Dreher et al. (2019) show that the region of birth of the leader receives more Chinese aid. They find no effect for World Bank aid. They also find no evidence of ethnic favoritism in either case. Compared to Dreher et al. (2019), we also find favoritism based on the leader's region of birth, but the magnitude is much larger. We also find evidence of geographic favoritism based on ethnic group ethnicity. We explain these differences by differential trends before and after treatment.[13]

Our finding on ethnic favoritism is relatively new in the literature. One notable exception is Perrotta Berlin et al. (2023). They find evidence of a lower allocation of World Bank aid to the ethnic region of the leader. This effect is attenuated when the country is a member of the UN Security Council. We find an opposite and very heterogeneous result with a different sample of countries: coethnic regions receive more World Bank aid. Our analysis differs in several ways. We focus on Africa, where ethnicity is more salient and democratic institutions are less widespread. We find that ethnic favoritism in World Bank aid disappears in democracies. We also observe favoritism for smaller geographic units, ADM2 rather than ADM1.[14]

More recently, Bomprezzi et al. (2024) find that bilateral aid from various sources is also directed to the birth regions of the president's wife. Finally, Bommer et al. (2022) find that U.S. aid after a natural disaster is disproportionately allocated to the president's birth region. They also find that favoritism is weaker in the presence of good institutions.

3. Data and variables

37 African countries make up our sample. They are divided into an average of 15 regions at the first administrative level (hereafter ADM1 regions) and 145 districts at the second administrative level (hereafter ADM2 districts).

We use three different data sources in this paper. (1) the list of presidents gives us the political transitions, their dates, and the ethnic group of the presidents (2) the list of aid projects gives us the date and location (ADM 1 and/or ADM 2) of aid projects in Africa. (3) the census (IPUMS), DHS or GREG ethnic map by order of preference gives us the ethnic composition of the administrative regions.

[8] This level of detail is rare for public spending or investment in low-income (and autocratic) countries, particularly in sub-Saharan Africa. This leaves most researchers with a choice between country case studies (André et al., 2018; Burgess et al., 2015) or replacing measures of investment with welfare outcomes (*e.g.*, infant mortality, educational attainment, or nightlight intensity, see De Luca et al. 2017, Franck and Rainer 2012, Kramon and Posner 2013, 2016, Perrotta Berlin 2013, Widmer and Zurlinden 2022). Measuring favoritism through welfare outcomes is indirect, and welfare outcomes could therefore be affected by unobserved factors (such as local economic conditions or specific shocks affecting a region).

[9] For example, Dreher et al. (2009) study the effect of UN Security Council membership on the number of World Bank projects in the country, and Alesina and Dollar (2000) study the pattern of foreign aid allocation and find that political alliances and the colonial past are important determinants of foreign aid. Kuziemko and Werker (2006) find that the amount of aid received by a country with a non-permanent seat on the UN Security Council increases by 59%. Hoeffler and Outram (2011) also find that aid allocation does not follow the needs of recipient countries, but rather the self-interest of donors.

[10] Foreign aid can lead to rent-seeking behavior (Reinikka & Svensson, 2004; Svensson, 2000) that weakens institutions and democracy (Bräutigam & Knack, 2004; Rajan & Subramanian, 2007)

[11] The World Bank has explicit goals of poverty reduction and shared prosperity (Bank, 2014) and applies principles of conditionality. We provide details on the anti-corruption, evaluation and transparency policies of the two donors in Appendix B.

[12] Even if this conditionality is imperfectly applied (see Hernandez, 2017; Kilby, 2009).

[13] See Appendix F

[14] In addition, we do not include areas that are coethnic with former presidents in the control group, which is important when comparing our results with Dreher et al. (2019).

3.1. List of presidents

Dreher et al. (2019) provide a list of African presidents from 2000 to 2011.[15] We complete their data between 1995 and 2014.[16][17] We make three exceptions, using prime ministers in Ethiopia, where the president is less powerful, and in the kingdoms of Eswatini (Swaziland) and Lesotho.[18] We code the ethnicity of politicians to ensure comparability with the geography of ethnic groups (Section 3.3). The detailed list of presidents (and their ethnicities) is reported in online Appendix Table H.5. We identify 86 presidential transitions between 1995 and 2014, with an average term of 5.07 years. Online Appendix Table H.1 lists these transitions.

We identify two types of transitions from the list of presidents. Ethnic transitions are characterized by a change in the president's ethnicity. Regional transitions are characterized by a change in the president's place of birth (at the ADM1 or ADM2 level depending on the specification). We identify 39 ethnic transitions between 1995 and 2014 with an average duration of 4.6 years between transitions, 67 regional transitions at the ADM1 administrative level and 69 at the ADM2 administrative level with average durations of 5.47 and 5.46 years between transitions, respectively. Online Appendix Tables H.2, H.3 and H.4 show the number of ethnic and regional transitions and their average duration.

3.2. Aid projects from China and the World Bank

Bluhm et al. (2018) provide impressive datasets describing the aid projects of China and the World Bank. The data are available from 2000 to 2011 for China and from 1995 to 2014 for the World Bank.[19][20] The Chinese aid data are collected from internet sources and may be subject to reporting bias. In both cases, geographic information is available with some missing data at the ADM1, ADM2 and/or geographic coordinate levels. We geolocate the geographic coordinates using the 3.6 version of the GADM data.[21] The list of World Bank projects includes 1982 projects in Africa. Of these, 1905 have some information on their geolocation: 826 are national, 209 are located at the ADM1 level and 870 are located at the ADM2 level. These 1079 projects with subnational locations generate a total of 15031 locations*projects. The data list 1955 Chinese aid projects in Africa, of which 1773 are geolocated: 925 are national, 69 provide information at the ADM1 level and 779 at the ADM2 level. A total of 848 projects with subnational geographic information generate 2034 locations*projects. We split projects with multiple localizations equally, as information on budget shares by region is not provided.

Throughout the paper, Y_{dt} is the amount of aid commitment for an ADM1 region or ADM2 district d in year t. We keep the total amounts, but the results remain almost identical when we exclude non-concessional loans.[22] Fig. 1 plots the annual commitments for China and the World Bank, respectively. Online Appendix Tables A.1 and A.2 show the sectoral distribution of Chinese and World Bank aid over time. They show that Chinese aid is highly concentrated in transport and energy infrastructure, while World Bank aid is diversified across many sectors.

We define y_{dt} as the inverse hyperbolic sine of Y_{dt}, which approximates $\log(Y_{dt})$.[23]

$$y_{dt} = \log^M(Y_{dt}) = \log\left(Y_{dt} + \sqrt{Y_{dt}^2 + 1}\right) \quad (1)$$

3.3. Ethnic groups

We use three sources of information to estimate the ethnic composition of the ADM1 and ADM2 regions: the IPUMS extract of the census, DHS data, and the GREG ethnic map (Weidmann, Rød, & Cederman, 2010). We create a coherent classification to ensure comparability between (Dreher et al., 2019) and the census/survey/GREG ethnic map, and we list an average of 13 ethnic groups per country, which are listed in online Appendix H.6. Online Appendix A.1 presents the detailed procedures.

Our ethnicity variable is E_{dp}; it measures the proportion of people from the ADM1 or ADM2 regions who share the ethnicity of president p (or the proportion of the area in countries where ethnicity is coded with GADM). $E'_{dp} = \mathbb{1}(E_{dp} \geq 50\%)$, is a discrete variable that takes the value 1 when the share of president p's the ethnic group is at least 50%. We call coethnic regions the ADM1 or ADM2 regions with $E'_{dp} = 1$.

Although each country is highly fragmented, we find that the ADM1 and especially the ADM2 regions are much more homogeneous. Of the 5323 ADM2 districts in the sample, 4033 (75%) have an ethnic majority of more than 50%. ADM1 regions are larger and therefore more fractionalized. Of the 558 ADM1 regions, 209 (37%) have no a strict ethnic majority. Ethnic favoritism should be more efficient and therefore more visible at the ADM2 level.

In online Appendix, Table A.3 presents summary statistics of the coethnicity variables, and Tables C.3 and C.4 show the robustness of our main results to alternative thresholds and to the use of the continuous coethnicity variable.

3.4. Difference between coethnicity with the president and region of birth of the president

In practice, the ADM1 birth region of newly appointed presidents and their coethnic districts do not coincide. Only 42% of the ADM2 districts of the ADM1 birth region are co-ethnic with the president (and 47% of the ADM2 birth districts). Only 8% of the coethnic districts are in the president's ADM1 birth region, and the ADM2 birth districts represent only 1% of the coethnic districts. The correlation between two dummies for "ADM2 district within ADM1 birth region" and "coethnic ADM2 district" is only 8% in our regression sample. Online Appendix Figure A.1 examines this. Panel A shows that the distribution of the share of ADM2 districts in the ADM1 birth region between presidential transitions is bimodal: typically, the birthplace is either fully coethnic or not. Similarly, Panel B shows that the coethnic share of the ADM2 district of birth is bimodal. This cannot be explained by presidents born in the national capital, but a few examples (such as Alassane Ouattara of Côte d'Ivoire or Joseph Kabila of DR Congo) point to a more general phenomenon of geographic mobility among elites. Finally, Panel C of online Appendix Figure A.1 shows that the ADM1 region of birth tends to account for only a small share of coethnic districts, although there are a few exceptions.

[15] We exclude Somalia, South Sudan, and small islands.

[16] We rely on several sources of information, including press articles about presidential elections. Appendix H describes the detailed methodology.

[17] The PLAD database (Bomprezzi et al., 2024) provides a more recent list of presidents. Our results remain similar when we exclude transitions that differ between the PLAD database and ours.

[18] The ethnicity and birthplace of kings tends to remain constant from generation to generation.

[19] We drop the year 2012 in the Chinese dataset, as in Dreher et al. (2019), because of doubts about the completeness of the projects from that year.

[20] Note that Bomprezzi et al. (2024) published recently a new database (godad) with more coverage of World Bank and Chinese aid projects, as well as 18 other donors.

[21] www.gadm.org

[22] We separate ODA from OOF for Chinese aid and IDA aid from IBRD loans for World Bank aid in the last columns of online Appendix Tables E.2 and E.3, respectively. Online Appendix Tables E.4 and E.5 reproduce our main Tables excluding non-concessional loans.

[23] Inverse hyperbolic sines allow a transformation similar to logarithms for regions that do not receive aid projects and give a value of y_{dt} equal to zero when $Y_{dt} = 0$. Recent papers using this transformation include (Bahar & Rapoport, 2018; Clemens & Tiongson, 2017; McKenzie, 2017).

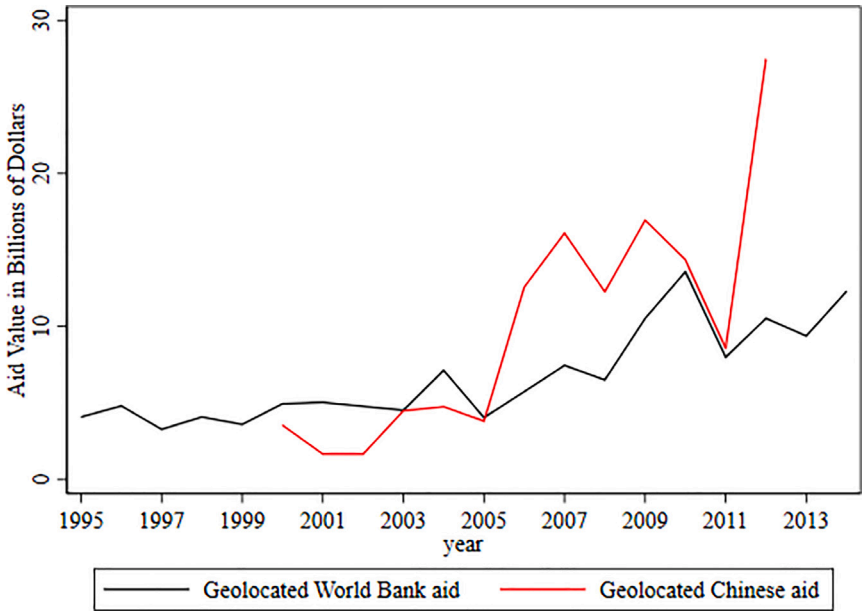


Fig. 1. Assistance projects values from World Bank and China. *Source:* Aiddata, our computations. Amounts in constant US Dollars.

4. Empirical strategy

We base our estimations on a comparison of the amount of aid projects before and after the appointment of a president, and between coethnic regions and other regions, or birth regions and other regions. This leads to the following equation for the differences in differences for ethnic favoritism:

$$n_{dtp} = \alpha E'_{dp}\mathcal{T} + \beta\mathcal{T} + \theta_{cp} + \eta_d + \varepsilon_{dtp} \quad \text{(DiD)}$$

where $\mathcal{T} = 1(t > T_p) + 0.5 * 1(t = T_p)$ takes value 1 when president p was appointed before year t, as T_p is the year of appointment of president p. It takes value 0.5 for the year of appointment, which we assume to be partially treated. Online Appendix Tables C.5 and C.6 provide robustness checks regarding the treatment of the year of appointment. Coefficient α is a difference in differences: between before and after year $T_p + 1$, and between districts d with a co-ethnicity with the appointed president p ($E_{dp} \geq 50\%$) and other districts. The country times transition fixed-effects θ_{cp} and region fixed effects η_d capture the effect of E'_{dp}.[24] We estimate (DiD) separately for each donor (Chinese or World Bank). We also study birth region targeting, using the same Eq. (DiD), only replacing E_{dp} with R_{dp}, a dummy taking value 1 in the president's region of birth.

Importantly, we include all transitions. We censor observations that are more than 10 years away from a transition: we keep observations if $T_p - 10 \leq t \leq T_p + 10$. Thus, if a year is close to several transitions, we include the corresponding observations of the same district d at time t several times for several transitions p. This duplication of observations could lead to an overestimation of the precision of our estimates. To avoid this, we cluster the standard errors of our estimates by ADM1 regions, as is commonly done in DiD and RDD settings. In the online appendix, we replicate our main specifications with bootstrapped standard errors (by ADM1 regions) and show that the standard errors are virtually identical, see online appendix Tables C.7 and C.8.[25]

The limitations of the more standard two-way fixed effects models are highlighted in a growing literature (Borusyak, Jaravel, & Spiess, 2024; Callaway & Sant'Anna, 2021; De Chaisemartin & d'Haultfoeuille, 2020; Sun & Abraham, 2021). We provide an intuitive way to overcome these problems, and show in online Appendix D that our results are robust to these more standard approaches. (See in particular the results in Tables D.1 and D.2). We reproduce our DiD results using tenure-adjusted windows (columns 1 to 4) and 5-year windows (columns 5 and 6) in online appendix Tables C.13 for Chinese aid and C.14 for World Bank aid. Finally, we run placebo regressions in online appendix Tables C.16 and C.17: we compare our main results (in columns 1 and 3 of these Tables) with regressions around other possible transition dates: executive elections[26] and transitions in which the leader's ethnicity or place of birth remain unchanged.

Comparing the variation in aid projects within each region around a transition date should capture region-specific factors (with region fixed-effects). In particular, this could capture ethnic background or colonial and pre-colonial history for instance.

Differences in differences have a well-known limitation: their sensitivity to the common trend assumption. Co-ethnic regions or the leader's birth region may have a pre-existing specific trend in aid projects. For example, economically dominant and dynamic ethnic groups may have a higher probability of having a co-ethnic president

[24] Including a region times transition fixed θ_{dp} gives very similar coefficient estimates. In another robustness check, we replace E'_{dp} with its continuous version E_{dp} and get very similar results (see online Appendix Tables C.3 and C.4).

[25] Recent literature revisits clustering practices and distinguishes between sampling-based and design-based clustering (Abadie, Athey, Imbens, & Wooldridge, 2020, 2023). Design-based clustering is relevant here: clustering is necessary because the treatment is correlated across observations. Athey and Imbens (2022) provide the treatment in DiD settings and show that clustering at the treatment unit is conservative. Online Appendix Tables C.9 and C.10 reproduce our main tables implementing this: clustering by ethnic group × transition units. Our results are robust to this change. In our view, clustering by ethnic group is irrelevant here: it would correspond to sample-based clustering, which means extrapolating our results to a larger population of ethnic groups. We cluster at the district level in the main body of the paper because this is standard in the literature and because, to the best of our knowledge, there is no treatment of design-based clustering for RDDs.

[26] We use the "National Elections across Democracy and Autocracy" dataset (see Hyde & Marinov, 2012)

(*e.g.*, more dynamic groups may invest more in competing for power). These more dynamic groups may also receive more aid to support their rapid economic development. Or the leader's region of birth may be a more urban and dense area with specific trends in development and aid. The DiD results may capture these district-specific trends rather than the direct effect of having a co-ethnic president or the president's birthplace. We use three methods to ensure the validity of our estimates. First, we rely on event studies to verify that there is no pre-existing trend and that the parallel trend assumption holds. Second, we extend (DiD) with a control for a difference in trends between the coethnic districts ($E'_{dt} = 1$) and the other districts (and/or a difference with the region of birth, if relevant). Third, we split these trends before and after the transition. This third specification becomes a refinement of Regression Discontinuity Designs: we compute regression discontinuities (in time) in coethnic districts and in other districts around transition dates, and we compute the difference in discontinuities between the two groups. This approach is sometimes called difference-in-discontinuities (Grembi, Nannicini, et al., 2012), or RD-DID.

$$n_{dtp} = \alpha E'_{dp}\mathcal{T} + \beta\mathcal{T} + P_{11}(t-T)\mathcal{T}E'_{dp} + P_{10}(t-T)(1-\mathcal{T})E'_{dp} + P_{01}(t-T)\mathcal{T}(1-E'_{dp}) + P_{00}(t-T)(1-\mathcal{T})(1-E'_{dp}) + \theta_{cp} + \eta_d + \varepsilon_{dtp} \quad \text{(RDD)}$$

In addition to the region and country times transition fixed-effects, we control for P_{00}, P_{01}, P_{10}, and P_{11}, four different trends in dates, for coethnic districts and for other districts (or between regions of birth and other regions), before and after the transition. Thus, β captures the discontinuity for non-coethnic districts, and α captures the additional discontinuity for coethnic districts. Thus, we have a difference between the coethnic and non-coethnic regions in the discontinuity RDD at the time of the political transition. Again, some RDD regressions use the continuous ethnicity variable E_{dp} instead of its binary version E'_{dp}. We also use the R_{dp} dummy to study birth region favoritism.

This specification probably solves most of the estimation problems. First, district-specific trends cannot bias our estimates, as this specification explicitly controls for a difference in trends in coethnic and non-coethnic districts. Second, the appointment of a new president may be endogenous. If there is a discontinuity in aid around the appointment date, it is unlikely to be related to other pre-existing factors that affect both aid and the appointment of a president from a particular ethnic group or region.[27]

We also provide estimates that simultaneously include the treatment for coethnic regions $E'_{dp}\mathcal{T}$ and the treatment for region of birth $R_{dp}\mathcal{T}$ (the explanatory variables are the reunion of the two specifications).

5. Empirical results

5.1. Main specifications

In this section, we present the results of the estimation of (DiD) and (RDD) for the two sources of aid and the two types of favoritism: ethnic and birthplace favoritism.

Table 1 shows the results for Chinese aid at the ADM1 administrative level. The corresponding table at the ADM2 level is presented in the online Appendix (Table C.1). Table 2 presents the results for World Bank aid at the ADM2 level (and online Appendix Table C.2 presents the ADM1 results).[28] In these tables, we examine ethnic favoritism separately in columns 1 and 3, and regional favoritism separately in columns 2 and 4. We study them together in columns 5 and 6. Columns 1 and 2 present the DiD specifications. Columns 3, 4, and 5 reproduce the same specifications, including specific trends for treated and control group regions. Column 6 reports coefficient estimates for the RDD specification. Importantly, we only include relevant transitions in columns 1 through 4: we keep ethnic transitions in columns 1 and 3,[29] and regional transitions in columns 2 and 4.[30] We keep ethnic and regional transitions in columns 5 and 6. In these columns, we recode $E'_{dp} = 0$ for transitions holding the president's ethnicity constant and $R_{dp} = 0$ for transitions holding the region of birth constant. Columns 7 and 8 analyze the extensive and intensive margins of favoritism, respectively. The extensive margin refers to the presence of aid in the ADM1 region or in the ADM2 district. The intensive margin refers to the amount of aid (conditional on receiving some aid). In columns 7 and 8, we keep regional transitions to study birthplace favoritism for Chinese aid, and we keep ethnic transitions to study ethnic favoritism for World Bank aid.

In every specification of Table 1, the birth regions of newly appointed presidents receive more Chinese aid after the appointment of a new president. This result is very robust and stable in magnitude across all specifications. The impact is extreme: the amounts in the birth region are roughly multiplied by $\exp(2.643) \approx 14$ compared to before the transition (and relative to the other regions). Thus, as already shown by Dreher et al. (2019) the magnitude of the effect of the president's birth region appears to be very large. Note, however, that the effect we find is much larger than in their study. This is because our specification excludes observations far from the transition dates.[31] We find that for an average transition, 26.2% of Chinese aid goes to the ADM1 region of the president's birth between 1 and 5 years after the transition. Our model explains ≈ 21 to 24 percentage points of this 26.2% by regional favoritism.[32] Online Appendix Table C.1 shows the corresponding result at the ADM2 level. The effect is much smaller and statistically insignificant. In other words, regional favoritism in Chinese aid does not necessarily target the president's exact birthplace, as measured by the ADM2 district but rather an extended area around her birthplace. This is likely due to the nature of Chinese aid. It disproportionately corresponds to large infrastructure and energy projects (about 65%, see online Appendix Table A.1 for summary statistics) and therefore necessarily targets larger geographic areas than ADM2 districts.

In column 7 of Table 1, we predict the probability that an ADM1 region receives some Chinese aid (regardless of the amount) using a linear probability model.[33] The dependent variable is the binary variable that takes the value of 1 if ADM1 region d receives a development project from China in year t. The coefficients α are positive and highly significant, indicating that the region of the president's birth increases the probability of receiving Chinese aid by 16.9 percentage points, with a sample average of 8%. In column 8, we examine the intensive margin, restricting the sample to the ADM1 regions times year that receive a (strictly) positive amount of Chinese aid. It reports the results

[27] Faye and Niehaus (2012) provide an example of US aid being manipulated to help friendly incumbent presidents before elections. This could happen within countries (*e.g.*, in districts that are coethnic with the incumbent). However, we look at the differences in aid between districts coethnic with the new president and other districts, and we find differences after the political turnover. (Aid given after the turnover cannot affect the turnover itself.)

[28] The ADM2 districts are more numerous than the ADM1 regions, which accounts for most of the difference in the number of observations between Tables 1 and 2. In addition, World Bank data are available between 1995 and 2014, while Chinese aid data are available between 2000 and 2011.

[29] Transitions in the president's ethnicity, see Section 3.1.

[30] Transitions in the president's place of birth, see Section 3.1

[31] Additional regressions show that including region-specific trends in Dreher et al. (2019)'s specifications can increase their coefficient to ≈ 1.7, while the coefficient in column 2 falls to 1.4 when we include region-specific trends. See online Appendix F for more details on this comparison between our results and Dreher et al. (2019). We thank Andreas Fuchs for providing the replication package.

[32] See Table 1 and especially Table note *a* for details.

[33] Online Appendix Table C.11 robustness using the marginal effects of a Probit model.

Table 1
The Effect of a Presidential Appointment on Chinese Aid (ADM1 level).

	dep.var.: $\log^M(Y_{dt})$, Y_{dt} Total Value of Chinese Aid						Extensive margin	Intensive margin
	DiD		Extended DiD			RDD	Extended DiD	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$E'_{dp} * \mathcal{T}$	0.269		0.568		0.160	0.388		
	(0.562)		(0.710)		(0.745)	(0.736)		
$R_{dp} * \mathcal{T}$		1.806**		2.643***	2.517**	2.452**	0.169***	−2.302*
		(0.717)		(0.933)	(1.001)	(1.031)	(0.059)	(1.255)
$\mathcal{T}$	0.570***	0.597***	−0.791**	−0.466**	−2.583**	−1.190	−0.025*	0.398
	(0.210)	(0.151)	(0.309)	(0.227)	(1.118)	(1.019)	(0.013)	(0.515)
E'_{dp} (Coethnic ADM1 District)	−0.053		0.220		0.036	−0.321		
	(0.145)		(0.547)		(0.520)	(0.449)		
R_{dp} (ADM1 District of Birth)		−0.697**		0.090	0.203	−1.001	0.010	−1.231
		(0.291)		(0.758)	(0.725)	(0.676)	(0.045)	(2.119)
$t - T_p$ (Linear Trend)			0.202***	0.157***	0.611***		0.009***	0.160**
			(0.034)	(0.026)	(0.134)		(0.001)	(0.063)
$(t - T_p) * E'_{dp}$			−0.043		−0.007			
			(0.079)		(0.072)			
$(t - T_p) * R_{dp}$				−0.115	−0.123		−0.008	0.221
				(0.097)	(0.096)		(0.006)	(0.212)
$(t - T_p)$ interacted with $E'_{dp}, R_{dp}, E'_{dp} * \mathcal{T}, R_{dp} * \mathcal{T}, \mathcal{T}, 1 - \mathcal{T}$						✓		
$\mathcal{T}$ and $(t - T_p)$ interacted with ethnic transition and with regional transition					✓	✓		
Country times Transition F.E.	✓	✓	✓	✓	✓	✓	✓	✓
Country times Regions F.E.	✓	✓	✓	✓	✓	✓	✓	✓
Observations close to an ethnic transition	✓		✓		✓	✓		
Observations close to a regional transition		✓		✓	✓	✓	✓	✓
N	4,657	6,433	4,657	6,433	6,757	6,757	6,433	453
Avg. share of aid in ADM1 of birth, regional transitions		0.262		0.262	0.262	0.262	0.165	
Avg. share of distorted aid[a] in ADM1 of birth, regional transitions		0.219		0.244	0.241	0.240	0.034	

Standard errors clustered at the ADM1 level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
[a] We simulate the potential aid by subtracting $\hat{\alpha} R_{dp} \mathcal{T}$ to the actual output. Distorted aid is the difference between actual and potential aid.

of OLS specifications using the $\log^M$ of the value of Chinese aid as the dependent variable. The coefficient α is negative and marginally significant.[34] In other words, the region of the president's birth is more likely to receive Chinese aid, but the conditional amounts are somewhat smaller on average.

Conversely, coethnic regions at the ADM1 and ADM2 levels do not attract more Chinese aid. Presidents seem to capture Chinese aid projects for their regions of origin rather than a broader targeting of all coethnic regions. Also, the coefficient on years after transition ($\mathcal{T}$) is positive in columns 1 and 2 of Table 1, but negative in columns 3 to 7, controlling for linear trends. The effect of a linear trend ($t - T_p$) is always positive, which explains this change.[35] The effect of the coethnic region dummy (E'_{dp}) is not statistically different from zero, and coethnic regions have a similar trend to other regions ($(t - T_p) * E'_{dp}$). Finally, the region of birth (R_{dp}) has a lower investment before the treatment ($\mathcal{T} = 0$) in column 2, but this effect disappears in columns 4 to 7 where we control for a different trend in the region of birth ($(t - T_p) * R_{dp}$). This last effect is not significant: the test fails to disentangle the two effects.

Fig. 2 graphically illustrates the results of Table 1 with sample averages. We plot the average $\log^M(Y_{dt})$ of Chinese aid as a function of time separately for the ADM1 region of the president's birth and other regions. This allows us to look at possible pre-trends. Consistent with the results of Table 1, where trends play a minor role, there is no visible pre-trend in Fig. 2 before the year of appointment. On the other hand, an increase in Chinese aid in the region of the president's birth is visible when $\mathcal{T} - t \geq 1$ and not apparent in other regions. While the estimates of the annual effects are imprecise, the average size of the effect is comparable to the coefficients in Table 1 (≈ 2 points).

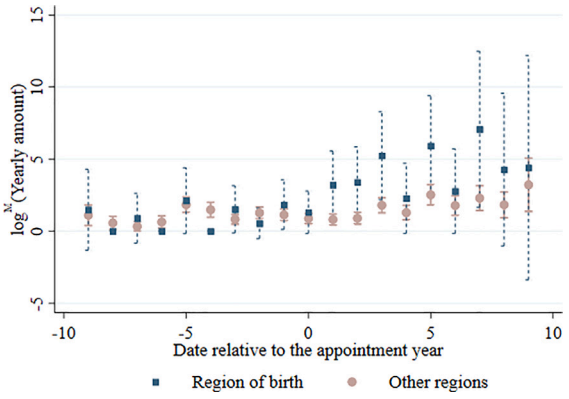


Fig. 2. Average yearly amount of Chinese aid around appointment years.
Notes: Date relative to the appointment year: difference between current year and appointment year ($t - \mathcal{T}$). Confidence intervals are obtained from an OLS regression of $\log^M(Y_{dt})$ on Date relative to the appointment year times region of birth dummies, clustered by ADM1 region.

[34] Online Appendix Table C.11 shows robustness using standard logs and Poisson regressions and number of projects.
[35] The nonlinearity behind the negative coefficient on $\mathcal{T}$ is more difficult to interpret.

Table 2
The Effect of a Presidential Appointment on World Bank aid at the ADM2 level.

	dep.var.: $\log^M(Y_{dt})$, Y_{dt} Total Value of World Bank Aid						Extensive margin	Intensive margin
	DiD		Extended DiD			RDD	Extended DiD	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$E'_{dp} * \mathcal{T}$	0.413**		0.681***		0.639**	0.867***	0.040**	0.725***
	(0.183)		(0.251)		(0.250)	(0.279)	(0.016)	(0.230)
$R_{dp} * \mathcal{T}$		0.077		0.947	1.596**	1.563**		
		(0.418)		(0.652)	(0.637)	(0.645)		
$\mathcal{T}$	0.409***	0.510***	−1.043***	−0.588***	2.144***	2.342***	−0.069***	0.296***
	(0.065)	(0.058)	(0.134)	(0.087)	(0.564)	(0.572)	(0.009)	(0.095)
E'_{dp} (Coethnic ADM1 District)	−0.250		−0.373**		−0.318**	−0.600**	−0.022**	−0.289**
	(0.171)		(0.165)		(0.137)	(0.250)	(0.011)	(0.126)
R_{dp} (ADM1 District of Birth)		0.299		−0.138	−0.442	−0.441		
		(0.215)		(0.361)	(0.373)	(0.488)		
$t - T_p$ (Linear Trend)			0.166***	0.123***	−0.026		0.011***	0.016*
			(0.016)	(0.013)	(0.023)		(0.001)	(0.009)
$(t - T_p) * E'_{dp}$			−0.031		−0.029		−0.001	−0.064***
			(0.033)		(0.032)		(0.002)	(0.023)
$(t - T_p) * R_{dp}$				−0.092	−0.179**			
				(0.086)	(0.083)			
$(t - T_p)$ interacted with $E'_{dp}, R_{dp}, E'_{dp} * \mathcal{T}, R_{dp} * \mathcal{T}, \mathcal{T}, 1 - \mathcal{T}$						✓		
$\mathcal{T}$ and $(t - T_p)$ interacted with ethnic transition and with regional transition					✓	✓		
Country times Transition F.E.	✓	✓	✓	✓	✓	✓	✓	✓
Country times Regions F.E.	✓	✓	✓	✓	✓	✓	✓	✓
Observations close to an ethnic transition	✓		✓		✓	✓	✓	✓
Observations close to a regional transition		✓		✓	✓	✓		
N	77,213	124,221	77,213	124,221	123,251	123,251	77,213	6,713
Avg. share of aid in ADM1 of birth, regional transitions	0.057		0.057		0.057	0.057	0.070	
Avg. share of distorted aid[a] in ADM1 of birth, regional transitions	0.019		0.028		0.027	0.033	0.016	

Standard errors clustered at the ADM1 level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
[a] We simulate the potential aid by subtracting $\hat{\alpha} R_{dp} \mathcal{T}$ to the actual output. Distorted aid is the difference between actual and potential aid.

The results for World Bank aid are very different. We present them in Table 2 at the ADM2 level and in online Appendix Table C.2 at the ADM1 level. We find evidence of ethnic favoritism at the ADM2 level. In each specification of Table 2 (ADM2), coethnic regions with a newly appointed president receive more aid projects. The magnitude of the coefficient is large but much smaller than for home region and Chinese aid. In our preferred specifications (columns 5 and 6), aid amounts in coethnic ADM2 districts are multiplied by $\exp(0.6) \approx 1.8$. We estimate that coethnic districts receive 5.7% of World Bank aid between 1 and 5 years after a turnover, and that about 3 percentage points of this 5.7% is due to ethnic favoritism.[36][37] In online Appendix Table C.2, the results are insignificant at the ADM1 level, suggesting that ethnic favoritism follows the fine geography of ethnicity rather than regional boundaries. Conversely, we find little evidence that favoritism in World Bank aid follows the president's region of birth. In Tables Table 2 and C.2, the president's place of birth receives slightly more aid projects only in columns 5 and 6 of Table 2,[38] but not in other specifications. Column 7 measures the extensive margin and reports the results of a linear probability model. The coefficient α is positive, indicating that coethnic districts increase the probability of receiving World Bank aid by 4.0 percentage points relative to other districts, with a sample average of 9%. Column 8 shows that the amounts are also larger conditional on receiving aid. Online Appendix Table C.12 shows that the results at the extensive margin are robust to the use of a probit model, and the result at the intensive margin is robust to the use of a standard log but not to the use of a Poisson regression.

These results are consistent with the World Bank's better control of clientelism. Ethnic favoritism is probably harder to detect because precise ethnic boundaries are not as well known as the president's region of birth. Clientelism accounts for a smaller share of total World Bank aid ($\approx 3\%$ versus $\approx 20\%$ for Chinese aid). In addition to a smaller magnitude, the patterns also seem less visible: ethnic boundaries at the ADM2 level are less obvious than in ADM1 regions. To our knowledge, we are the first paper to highlight elite capture in the form of ethnic favoritism in World Bank aid.

The coefficient on years after the transition ($\mathcal{T}$) is positive in columns 1 and 2 of Table 2, but becomes negative in columns 3, 4, and 7, controlling for linear trends. The effect of a linear trend $(t - T_p)$ is usually positive and explains this change.[39] Columns 5 and 6 are

[36] See Table 2 and especially Table note *a* for details.
[37] We compare our results with the working paper version of Dreher et al. (2019), Dreher et al. (2016), using the data from Dreher et al. (2019). The main difference is that the two-way fixed effects specification of Dreher et al. (2019) considers former co-ethnic districts as having lost their treatment, while we consider them as controls. Indeed, dynamic effects due to the pipeline of projects are possible: ethnic districts of former presidents may retain their advantages a few years after the transition. Therefore, assigning a treatment of 0 to formerly treated regions may be a strong assumption: dropping previously treated observations from Dreher et al. (2019)'s makes their results quantitatively close to ours (and does not change the results for our sample). See online Appendix F for details.
[38] Further investigation shows that controlling for coethnic-districts specific trends increases the coefficient of birthplace favoritism.
[39] The nonlinearity behind the negative coefficient for $\mathcal{T}$ is more difficult to interpret.

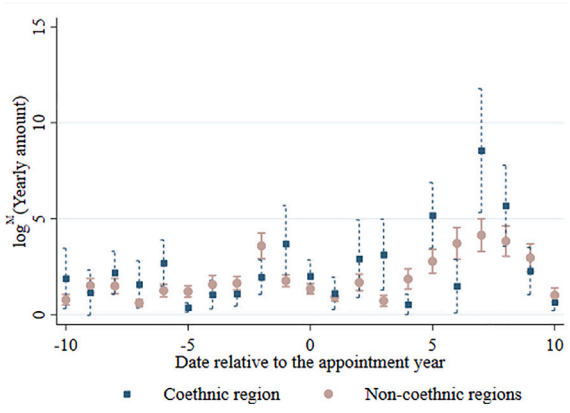


Fig. 3. Average yearly amount of World Bank aid around appointment years, sample with World Bank investment in the country the year before the transition.
Notes: Date relative to the appointment year: difference between current year and appointment year ($t - \mathcal{T}$). Confidence intervals are obtained from an OLS regression of $\log^M(Y_{dt})$ on Date relative to the appointment year times dummies for coethnic ADM2 regions, clustered by ADM1 region.

exceptions, where the increase in World Bank aid over time is driven by a positive coefficient on $\mathcal{T}$. The effect of the coethnic district dummy (E'_{dp}) is negative and usually significant: coethnic districts receive less World Bank aid before treatment (when $\mathcal{T} = 0$). Coethnic districts also have a similar trend to other regions ($(t - T_p) * E'_{dp}$), except in column 8. Finally, the district of birth (R_{dp}) receives a similar amount before treatment (when $\mathcal{T} = 0$) and the trend in the region of birth ($(t - T_p) * R_{dp}$) may be smaller, although this is only significant in one specification.

Fig. 3 graphically illustrates the results from Table 2 with sample averages. Since the treatment effect is heterogeneous, we focus on a group with a higher treatment effect, the countries with some World Bank aid before the presidential change. The results using the full sample are shown in online Appendix Figure A.2 and are more noisy. We plot the average $\log^M(Y_{dt})$ of World Bank aid as a function of time separately for coethnic regions and other regions. This allows a closer look at possible pre-trends. Trends seem to play a minor role in Table 2, but they are not visible in Fig. 3.[40][41] On the other hand, coethnic districts seem to receive slightly more aid than other regions when $\mathcal{T} - t \geq 1$ in Fig. 3. This is not always the case (see $t = 4$ or $t = 6$), but we consider this likely to be random. Finally, Fig. 3 shows the average trend in non-coethnic regions. The average trend of increasing amounts over time is visible and explains the positive coefficients of $\mathcal{T}$ in columns 1 and 2 of Table 2.

5.2. When and where does favoritism occur ?

Tables 3 and 4 present the heterogeneity of our main effects. We examine heterogeneous effects with respect to institutions, interactions between Chinese aid and World Bank aid, and past relations between the donor and recipient countries.

Favoritism in democracy and dictatorship

We examine the role of democracy for the World Bank aid in Table 3. Column 1 of Table 3 introduces an interaction between the ethnic favoritism treatment variable and dictatorship. We define a dictatorship as a country with a Polity 2 (Marshall & Gurr, 2020) score below 0 in the year following the presidential transition. This allows us to compare the pattern of ethnic favoritism between democracy and dictatorship. We have 33 changes in the president's ethnic group in democracy and 6 in dictatorship. Table 3, column 1 shows that ethnic favoritism in World Bank aid is much larger in dictatorship. Note, however, that democracy does not necessarily always prevent ethnic favoritism. For example, De Luca et al. (2017) find some evidence of ethnically targeted policies by looking at nightlight intensity around the world, even in established democracies and developed economies, while Mueller and Tapsoba (2016), André et al. (2018) and Burgess et al. (2015) find that ethnic favoritism disappears when a country becomes democratic.

The Polity 2 consists of two sub-indexes: an index of democracy and an index of autocracy, which we test separately in columns 2 and 3 of Table 3. These columns clearly show that the effect is non-linear: the index of autocracy has a stronger effect than the effect of democracy. World Bank favoritism is amplified in the strongest autocracies, but not reduced in stronger democracies (relative to intermediate situations).

Table 3, column 4 shows the results when the autocracy index is split: the two crucial subcomponents that affect the pattern of favoritism are the degree of constraints on the executive and the fact that the recent transition was unregulated (the president was neither elected nor nominated nor interim.). Online Appendix Table C.15 shows the detailed results with all Polity variables: the differences among the regulated selection of the head of the executive (*e.g.* election or nomination) or the degree of competitiveness of the selection process do not significantly affect World Bank aid favoritism. Note that column 4 shows that the flow of World Bank aid to co-ethnic areas after an undemocratic transition is relative to the country average. In fact, compared to more democratic countries, countries in which the executive has more authority and countries where the change of leadership was not regulated lose World Bank aid after a presidential change. (The coefficients are −1.489 and −2.908, respectively). Co-ethnic districts are just protected from this phenomenon.

In Table 3, columns 5, 6, and 7, we assess the robustness of our findings on democracy controlling for other institutional variables from the Quality of Government database (Teorell et al., 2018). We focus here on results comparing democracy to two other concepts: control of corruption and state capacity. While these concepts are correlated with democracy in practice, they have the advantage of having very different theoretical foundations.[42]

Table 3, column 5, tests several corruption indices, and shows no evidence that the level of corruption changes World Bank favoritism: the coefficients and the joint F-test on these coefficients fail to reject the null. Columns 6 and 7 test several indices of state capacity. Column 7 adds another index that is not available after 2011, resulting in a smaller sample for this regression. The results for state capacity are mixed: a coefficient in column 6 is marginally significant: there may be more World Bank favoritism when regulatory quality is higher. However, this coefficient loses significance in column 7, where legal institutional quality is positively correlated with World Bank favoritism. This time, the coefficient is negative: there is less favoritism when legal institutional quality is higher. However, these variables do not change the sign, significance, and magnitude of the effect of democracy on World Bank favoritism.

Column 1 of Table 4 introduces an interaction between the regional favoritism for Chinese aid and dictatorship (still defined by a Polity 2 score below 0 in the year following the presidential transition). We

[40] Additional regressions would show that this is due to the fixed effects of Table 2.

[41] The point for $t - \mathcal{T} = -1$ could indicate a short pre-trend. We believe this should be discarded because the annual estimates are highly imprecise; in particular, the point for $t - \mathcal{T} = -2$ has the opposite sign with a more precise estimated coefficient.

[42] Additional results showing all the variables of the Quality of Government database we tested are available upon request. Interestingly, using the democracy variables from Cheibub, Gandhi, and Vreeland (2010) or Geddes, Wright, and Frantz (2014) tends to confirm our results.

Table 3
The Effect of a Presidential Appointment on World Bank aid, democracy indicators.

	dep.var.: $\log^M(Y_{dt})$, Y_{dt} Total Value of World Bank Aid						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$E'_{dp} * \mathcal{T}$	0.556**	1.318	0.498*	−0.186	−0.413	−0.025	6.650*
	(0.239)	(1.234)	(0.259)	(0.306)	(3.139)	(0.634)	(3.399)
$E'_{dp} * \mathcal{T} *$ Dictatorship after the transition	2.471***						
	(0.756)						
$E'_{dp} * \mathcal{T} *$ Democracy Polity sub-index		−0.141					
		(0.198)					
$E'_{dp} * \mathcal{T} *$ Autocracy Polity sub-index			0.545**				
			(0.242)				
$E'_{dp} * \mathcal{T} *$ Executive has more authority than any accountability group[a]				1.233**	1.698***	1.955***	1.427**
				(0.514)	(0.614)	(0.621)	(0.550)
$E'_{dp} * \mathcal{T} *$ Unregulated change in the executive				2.901***	3.473***	3.842***	3.653***
				(0.633)	(0.563)	(0.670)	(0.805)
$\mathcal{T} *$ Executive has more authority than any accountability group				−1.489***	−1.469***	−0.964***	−1.998***
				(0.238)	(0.247)	(0.229)	(0.282)
$\mathcal{T} *$ Unregulated change in the executive				−2.908***	−2.624***	−3.449***	−3.268***
				(0.513)	(0.482)	(0.517)	(0.583)
$E'_{dp} * \mathcal{T} *$ Bayesian Corruption Indicator[b]					0.033		
					(0.066)		
$E'_{dp} * \mathcal{T} *$ V-Dem Political corruption index[c]					−1.682		
					(3.081)		
$E'_{dp} * \mathcal{T} *$ Worldwide Governance Indicators Control of Corruption[d]					0.915		
					(1.341)		
$E'_{dp} * \mathcal{T} *$ Worldwide Governance Indicators Government Effectiveness[d]						−1.483	−0.057
						(1.205)	(1.874)
$E'_{dp} * \mathcal{T} *$ Worldwide Governance Indicators Regulatory Quality[d]						2.474*	3.193
						(1.419)	(2.121)
$E'_{dp} * \mathcal{T} *$ Hanson and Sigman State Capacity Index[e]						1.338	−1.287
						(0.908)	(0.969)
$E'_{dp} * \mathcal{T} *$ Legal institutional quality, Institutional Quality Dataset[f]							−13.038**
							(5.645)
Controls present in column 3 of Table 2	✓	✓	✓	✓	✓	✓	✓
Controls interacted with dictatorship	✓						
Controls interacted with Polity Subindex		✓	✓				
Controls interacted with Polity Variables				✓			
Controls interacted with Corruption Indices					✓		
Controls interacted with State Capacity Indices						✓	✓
N	77,213	71,650	71,650	71,650	71,650	71,650	55,051
F-test: $E'_{dp} * \mathcal{T} *$ all corruption or state capacity indices					1.151	2.467	2.708
p-value					0.329	0.062	0.031

Standard errors clustered at the ADM1 level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[a] Takes value 1 when the *xconst* component of Polity 4 is 5 or less (see Marshall & Gurr, 2020).

[b] The Bayesian Corruption Indicator is created by Standaert (2015).

[c] Political corruption index from the Varieties of Democracy (V-Dem) Project (Pemstein et al., 2022).

[d] The Worldwide Governance Indicators: Kaufmann, Kraay, and Mastruzzi (2010).

[e] See Hanson and Sigman (2021).

[f] See Kunčič (2014).

have 50 changes in the president's birthplace in democracies and 17 in dictatorships. Table 4, column 2 shows that regional favoritism for Chinese aid seems to have a similar magnitude in democracy and in autocracy, the interaction term is close to zero and not statistically significant.

The differences in heterogeneity between World Bank and Chinese aid are interesting. Democratization may make ethnicity less salient in politics because leaders target larger segments of the population when institutions are more inclusive. However, favoritism in Chinese aid does not diminish with democracy: not all forms of favoritism disappear in democracy. The government's incentives to provide some public goods to the population and the donor's willingness to prevent corruption may complement each other to avoid favoritism: democratic checks and balances may only limit ethnic favoritism in World Bank aid allocation.

Do the World Bank and China influence each other?

In this section, we examine whether the amount of Chinese aid affects World Bank favoritism and, conversely whether the amount of World Bank aid affects Chinese aid favoritism. To examine this issue, we include an interaction term between our treatment variable and aid from the alternative source in columns 2 and 5 of Table 4. We consider two measures in our interaction. First, total aid from the other donor in the year before the transition. Second, total aid at the national level in the year before the transition. The idea is to understand whether the

Table 4
Heterogeneity in favoritism on Chinese and World Bank aid.

	Sample: turnovers between 2001 and 2011: Regional transitions, ADM1 level			Ethnic transitions, ADM2 level		
	$\log^M(Y_{dt})$, Y_{dt} = Value of Chinese Aid			$\log^M(Y_{dt})$, Y_{dt} = Value of World Bank Aid		
	(1)	(2)	(3)	(4)	(5)	(6)
$R_{dp} * \mathcal{T}$	2.810**	3.556***	2.939***			
	(1.106)	(1.334)	(1.046)			
$E'_{dp} * \mathcal{T}$				0.865**	−0.290	0.772**
				(0.351)	(0.573)	(0.342)
$R_{dp} * \mathcal{T}$ * dictatorship after the transition	−0.294					
	(1.860)					
$R_{dp} * \mathcal{T} * \log^M(Y^{WB}_{(T-1)})$[a]		−2.343				
		(2.036)				
$R_{dp} * \mathcal{T} * \log^M(Y^{WB}_{c(T-1)})$[b]		−0.099				
		(0.093)				
$R_{dp} * \mathcal{T} * \log^M(Y^{CH}_{(T-1)})$[c]			−0.343			
			(0.883)			
$R_{dp} * \mathcal{T} * \log^M(Y^{CH}_{c(T-1)})$[d]			−0.077			
			(0.097)			
$E'_{dp} * \mathcal{T} * \log^M(Y^{WB}_{(T-1)})$[c]					1.370**	
					(0.652)	
$E'_{dp} * \mathcal{T} * \log^M(Y^{WB}_{c(T-1)})$[d]					0.091**	
					(0.036)	
$E'_{dp} * \mathcal{T} * \log^M(Y^{CH}_{(T-1)})$[a]						1.454***
						(0.488)
$E'_{dp} * \mathcal{T} * \log^M(Y^{CH}_{c(T-1)})$[b]						0.000
						(0.038)
Country times Transition F.E.	✓	✓	✓	✓	✓	✓
Country times Regions F.E.	✓	✓	✓	✓	✓	✓
Controls: Dummies for region of birth and after the transition, linear trends in the region of birth and other regions	✓	✓	✓			
Controls: Dummies for coethnic district and after the transition, linear trends in coethnic and non-coethnic districts				✓	✓	✓
Controls interacted with dictatorship	✓					
Controls interacted with $\log^M(Y^{WB}_{(T-1)})$ and $\log^M(Y^{WB}_{c(T-1)})$		✓				✓
Controls interacted with $\log^M(Y^{CH}_{(T-1)})$ and $\log^M(Y^{CH}_{c(T-1)})$			✓		✓	
N	6,433	6,433	6,433	45,460	45,460	45,460

Standard errors clustered at the ADM1 level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[a] $Y^{WB}_{(T-1)}$ is the World Bank aid the year before the transition. $log^M(Y^{WB}_{(T-1)})$ is centered.

[b] $Y^{WB}_{c(T-1)}$ is the World Bank in the country the year before the transition.

[c] $Y^{CH}_{(T-1)}$ is the Chinese aid the year before the transition. $log^M(Y^{CH}_{(T-1)})$ is centered.

[d] $Y^{CH}_{c(T-1)}$ is the Chinese aid in the country the year before the transition.

pattern of favoritism in aid from one source is favored by the arrival of aid from another source, both at the global level and at the country level. Large Chinese aid flows may compete with other donors such as the World Bank, which could undermine its control over the political capture of aid.

Table 4, column 2 finds that Chinese aid favoritism is not affected by World Bank aid. Table 4, column 4 reproduces the results of Table 2, column 3 for the smaller sample of transitions between 2001 and 2011 for which we have data on Chinese aid. Column 4 examines the effect of Chinese aid on World Bank favoritism. It shows that World Bank aid favoritism is more intense in the years when Chinese aid is more intense. However, World Bank aid favoritism is not more intense in countries with more Chinese presence. This effect does not exist when examining comparisons between countries, making endogeneity concerns more likely. Nevertheless, it suggests that competition from Chinese aid may increase World Bank aid and favoritism.

Intensity of the aid flows and heterogeneity

Columns 2, 3 and 6 of Table 4 test whether aid favoritism is driven by the amount of aid from the same donor: the World Bank in column 3 and China in column 6. Similar to column 2, it tests whether the ($\log^M$ of) total annual amount before the transition has an effect and whether the country's annual amount before the transition has an effect. Column 3 of Table 4 finds no effect of the intensity of Chinese flows on Chinese aid favoritism. On the other hand, column 6 of Table 4 shows that World Bank aid favoritism is more intense when its aid is larger before the transition and in countries with more World Bank aid before the transition. The second coefficient is interesting because it suggests that pre-existing relationships may increase favoritism.

In our view, this finding suggests that the World Bank loses some bargaining power vis-à-vis national governments when pre-existing relations are more intense. This could be due to the costs of limiting aid in countries with good relations with the World Bank, for example for some actors within the Bank. This is not necessarily the case for Chinese

aid: if Chinese policies do not limit favoritism, we do not expect Chinese bargaining power to reduce favoritism. The first coefficient says that favoritism is more intense in years when World Bank aid is more intense globally. There are probably many interpretations of this. It could mean that larger amounts of aid globally may limit the World Bank's ability to control total amounts, but there are probably other interpretations as well.

Other heterogeneities

Online Appendix E estimates some other dimensions of heterogeneity in our coefficients. Table E.1 reports rural–urban heterogeneity. Birthplace favoritism for Chinese aid occurs mostly in urban areas, probably because Chinese aid tends to be concentrated in urban centers. The magnitude of World Bank aid favoritism appears to be similar between urban and rural areas.

Tables E.2 and E.3 show the heterogeneity of our results by type of aid. In both cases, we find that favoritism is more intense for aid flows (ODA for Chinese aid, IDA for World Bank aid), and insignificant for loans (other official flows for Chinese aid, IBRD loans for World Bank aid). Tables E.2 and E.3 also analyze regional and ethnic favoritism in the different sectors. Table E.2 shows that regional favoritism in Chinese aid seems to be visible in the sector where China is most active, such as transport, but also in health or social protection. In Table E.3, we find evidence of ethnic favoritism in health, transport, and agriculture projects.

6. Conclusion

In this paper, we use difference-in-differences and regression discontinuity designs to estimate the effect of regional and ethnic favoritism in the allocation of Chinese and World Bank aid. Specifically, we test whether a change in the president's ethnic group or region of birth can affect the amount of aid an ADM1 or ADM2 region receives. We find no evidence of ethnic favoritism in the allocation of Chinese aid and no evidence of regional and ethnic favoritism in the allocation of World Bank aid within ADM1 regions. However, an ADM1 region receives more Chinese aid if it is the birth region of the president. The ADM2 districts that share the president's ethnicity receive more World Bank aid than the other districts.

The magnitude of these effects is very large for Chinese aid compared to World Bank aid. Because the president's birthplace is well-known, it is difficult for recipient government officials to direct donor-funded projects to that location if the donor wishes to avoid favoritism or the appearance of favoritism. However, it is harder for donors to identify coethnic districts, making it an easier form of favoritism when donors are trying to control favoritism. Indeed, Chinese aid is notoriously less monitored than World Bank aid, which has more explicit development objectives. Therefore, presidents can target Chinese aid to their regions of birth and may be forced to target coethnic districts with World Bank aid.

We also show that ethnic favoritism in the allocation of World Bank aid disappears after democratization, is stronger when the Chinese aid flows are more intense, and is stronger in countries with more pre-existing aid relations with the World Bank are more intense. This suggests that democratic checks and balances in the host country, together with donor scrutiny, can prevent ethnic favoritism in the allocation of World Bank aid, at least in some circumstances.

These results provide new insights into aid allocation and, more generally, favoritism in policy implementation. First, they highlight that while both types of favoritism coexist in the political economy literature, they appear to occur in different contexts. Birth regions targeting is more visible and likely to occur in contexts where there are weaker checks and balances on policy. The targeting of co-ethnic regions, which is more diffuse and less visible, occurs in contexts where external controls and constraints on policy are stronger. Our results suggest that the two forms of favoritism are likely substitutes, as we do not observe them simultaneously in both sources of aid. It would be interesting to extend this line of research to understand in other contexts which types of policies and contexts allow for targeting of birth regions or, alternatively, co-ethnic regions.

Our results also suggest that the institutional context, competition between donors, or the longevity of relationships with the donor affect favoritism in aid allocation. These findings have important implications from a donor perspective, as they suggest that many situations are conducive to capture of aid for political purposes, which calls into question the development objectives of aid, especially from the World Bank. They imply that donors are likely to need to tailor their controls to the context in which aid is delivered, bearing in mind that certain contexts, such as the presence of other donors or weak institutions, may require increased controls.

CRediT authorship contribution statement

Pierre André: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation. **Paul Maarek:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Conceptualization. **Fatoumata Tapo:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation.

Declaration of competing interest

We have received no financial support for this research beyond that provided by our respective universities for travel to conferences, so we can unequivocally state that we have no conflicts of interest.

We do not have any other conflict of interest to declare. We confirm that this manuscript has not been published elsewhere and is not under consideration by another journal.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.worlddev.2024.106811.

Data availability

The replication files will be available at: https://doi.org/10.7910/DVN/EL6Z3X.

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